Lecture 1

- Maxwell's equations
  - Two forms
- Wave equation.
- Plane wave soln.
- Orthogonality cond.
- Energy and Intensity

At the end of the day, everything is derived from Maxwell's equations:

Every physicist must know these by heart: Свалад zpaaлb :)

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon} \quad \text{Gauss's law}$$

$$\vec{\nabla}\cdot\vec{B} = 0 \quad \text{no magnetic monopole}$$

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \quad \text{Faraday's law}$$

$$\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t} \quad \text{Ampere's law + Maxwell's condition.}$$

Divergence theorem states

$$\int_V \vec{\nabla}\cdot\vec{F}\,dv = \int_S \vec{F}\cdot d\vec{s}$$

and Stokes theorem $\int_C \vec{F}\cdot d\vec{r} = \int_S (\vec{\nabla}\times\vec{F})\,d\vec{s}$, where C encloses S.

Always have a mental picture in mind when talking about these theorems.
So, Integral formulation of Maxwell's equations is

$$\oint_V \vec{\nabla}\cdot\vec{E}\,dV = \oint_S \vec{E}\cdot d\vec{s} = \frac{1}{\varepsilon_0}\oint_V \rho\,dv = \frac{Q_{total}}{\varepsilon_0}$$

$$\oint_S \vec{B}\cdot d\vec{s} = 0$$

$$\oint_S (\vec{\nabla}\times\vec{E})\,d\vec{s} = \oint_C \vec{E}\cdot d\vec{l} = \oint_S -\frac{\partial\vec{B}}{\partial t}\cdot d\vec{s} = \frac{\partial}{\partial t}\Phi_{B,S}$$

flux through a surface.

$$= \oint_C \vec{B}\cdot d\vec{l} = \oint_S \mu_0\vec{J}\cdot d\vec{s} + \oint_S \mu_0\varepsilon_0\cdot\frac{\partial\vec{E}}{\partial t}\cdot d\vec{s} = \mu_0 I_S + \mu_0\varepsilon_0\cdot\frac{\partial}{\partial t}\Phi_{E,S}$$

By defining $\vec{D} = \varepsilon_0\vec{E}$ and $\vec{H} = \vec{B}/\mu_0$ and assuming empty space:

$$\vec{\nabla}\cdot\vec{D} = 0 \qquad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\cdot\vec{B} = 0 \qquad \vec{\nabla}\times\vec{H} = \frac{\partial\vec{D}}{\partial t}$$

In some medium, $\vec{D} = \varepsilon_0\vec{E} + \vec{P}$, $\vec{H} = \vec{B}/\mu_0 - \vec{M}$

applied external fields

responses of the medium.

In isotropic and linear medium, we have: $\vec{D} = \varepsilon\vec{E}$ and $\vec{B} = \mu\vec{H}$

note that we do not have x_0 subscripts.

Wave Equation

$$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}\times\left(-\frac{\partial\vec{B}}{\partial t}\right)$$

Vector Identity $= \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \vec{\nabla}^2\vec{E}$

is zero

$$= -\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B})$$

$$= -\frac{\partial}{\partial t}\left(\mu\varepsilon\frac{\partial\vec{E}}{\partial t}\right)$$

$$= -\mu\varepsilon\cdot\frac{\partial^2\vec{E}}{\partial t^2}$$

Note that $\vec{\nabla}^2\vec{E} \neq \vec{\nabla}(\vec{\nabla}\cdot\vec{E})$

Laplacian

$$\vec{E} = E_x(x,y,z)\hat{x} + E_y(x,y,z)\hat{y} + E_z(x,y,z)\cdot\hat{z}.$$

$$\vec{\nabla}^2\vec{E} = \frac{\partial^2 E_x}{\partial x^2}\hat{x} + \frac{\partial^2 E_y}{\partial y^2}\hat{y} + \frac{\partial^2 E_z}{\partial z^2}\hat{z}$$

$$\vec{\nabla}\cdot\vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \phi \text{ (scalar field)}$$

$$\vec{\nabla}(\vec{\nabla}\cdot\vec{E}) = \frac{\partial\phi}{\partial x}\hat{x} + \frac{\partial\phi}{\partial y}\hat{y} + \frac{\partial\phi}{\partial z}\hat{z}$$

are not necessarily equal

Overall, we get

$$\boxed{\vec\nabla^2\cdot\vec E - \mu\cdot\varepsilon\cdot\frac{\partial^2\vec E}{\partial t^2} = 0}$$

We can repeat this process for $\vec B$.

$$\vec\nabla\times(\vec\nabla\times\vec B) = \mu\cdot\varepsilon\cdot\frac{\partial}{\partial t}(\vec\nabla\times\vec E)$$

$$-\vec\nabla^2\cdot\vec B = \mu\cdot\varepsilon\cdot\frac{\partial}{\partial t}\left(-\frac{\partial\vec B}{\partial t}\right)$$

$$\boxed{\vec\nabla^2\cdot\vec B - \mu\varepsilon\frac{\partial^2\vec B}{\partial t^2} = 0}$$

One possible solution is a plane wave equation:

$$\vec E = \vec E_0\cdot e^{-i(\vec K\cdot\vec r - \omega t + \varphi)}$$

Check that this is a solution.

$\vec K$ - wave vector, energy propagation direction.

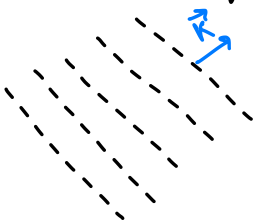
$K = |\vec K| = 2\pi/\lambda$ and $\omega = 2\pi f$

Dispersion relation: $\omega = cK/n$

How do lines of constant phase look like?

$\vec K\cdot\vec r - \omega t = \phi_0$ or at some time $t \Rightarrow \vec K\cdot\vec r = \text{const}$

This requires $\vec K\cdot\vec r = 0$! or $\vec K\perp\vec r$.



One can show that we must have

$$\boxed{\begin{aligned}\vec K\cdot\vec E &= 0\\ \vec K\cdot\vec B &= 0\\ \omega\vec B &= \vec K\times\vec E\\ |\vec E| &= \frac{c}{n}|\vec B|\end{aligned}}$$

mutually orthogonal $\vec E$ and $\vec B$ move along $\vec K$.

Energy and Intensity.

$$W = \vec F\cdot d\vec r$$
$$= q(\vec E + \vec v\times\vec B)\cdot\vec v\cdot dt$$
$$= q\,\vec E\cdot\vec v\cdot dt$$

For some charge distribution $q = \rho dV$, and

$$\frac{W}{dt\cdot dV} = \vec E\cdot\rho\vec v = \vec E\cdot\vec J$$

Power per unit volume.

"power dissipation comes from moving charges in $\vec E$ field."

Now, some derivational leap that you have to check:

$$-\vec{E}\cdot\vec{J} = \frac{\partial u}{\partial t} + \vec{\nabla}\cdot\vec{S}$$

$u = \frac{1}{2}(\varepsilon E^2 + B^2/\mu) \rightarrow$ energy density

$\vec{S} = \vec{E}\times\vec{B}/\mu \rightarrow$ Poynting vector.

With no free currents $\Rightarrow$ $\boxed{\frac{\partial \vec{u}}{\partial t} = -\vec{\nabla}\cdot\vec{S}}$ Energy conservation.

For some reason, $I = \langle|\vec{S}|\rangle = \frac{1}{2}\varepsilon_0 c|\vec{E}|^2 \propto |\vec{E}|^2$.